Paper VI — Weyl–Matter Cross Response in the BPB/MBE Framework

Reconciling ACT Lensing and E_G through Distinct Auto- and Cross-Weyl Sectors

v0.2.1 corpus-aware source release

F. Pelletier

2026

Abstract

We extend the BPB/MBE observational sequence to the E_G gravity statistic, a compressed probe of the relation between Weyl lensing, galaxy clustering, and velocity growth. Papers 0–V established the strict BPB/MBE dictionary and its first observational descendants across low-redshift expansion/growth, active-galaxy dynamics, CMB/Weyl consistency, neutron-star tidal structure, and early-galaxy formation. Paper VI addresses the tension exposed by those gates: ACT lensing prefers a positive Weyl auto-response, whereas E_G probes a lower Weyl–matter cross response. The resolution is not to force a single universal Σ. ACT lensing measures a Weyl auto-power sector, schematically $P_{WW} = \Sigma^2_{\rm auto} P_{\delta\delta}$, while E_G probes $P_{W\delta} = r_{W\delta}\Sigma_{\rm auto} P_{\delta\delta}$. The compressed v0.1 diagnostic gives $r^{\rm free}_{W\delta} = 0.633563$ and $\chi^2_{E_G} = 0.191532$ for ACTDR6+PlanckPR4+BOSS compressed bins. The v0.2 upgrade adds the projected density-weighted closure $r_{W\delta} = \sqrt{a_{\rm bg}} = 0.635364421415$, where $a_{\rm bg} = 0.403687948$ is inherited from the BPB background dictionary, not fit to E_G. The fixed closure is essentially indistinguishable from the free diagnostic: the recorded fixed-minus-free penalty is $\Delta\chi^2 = 0.008545$, while the amplitude-space support gives $r^2_{W\delta} = 0.397990327336$, only -1.411392% from $a_{\rm bg}$. We also retain the v0.1 corrected scale diagnostic: LOWZ probes $k \simeq 0.037$–$0.181\,{\rm Mpc}^{-1}$ and CMASS probes $k \simeq 0.023$–$0.198\,{\rm Mpc}^{-1}$ under $k = \ell/\chi$, so no bin lies in the ultra-low-$k < 0.02\,{\rm Mpc}^{-1}$ regime. The allowed claim is therefore structural: BPB/MBE requires $\mu_m \neq \Sigma_{\rm auto} \neq \Sigma_{\rm cross}$. The guardrail is explicit: this is not yet a full projected $C_\ell^{\kappa g}$, C_ℓ^{gg}, RSD, galaxy-bias, and covariance likelihood, nor a completed action-level perturbation proof of the density-weighted mixing relation.

1 Introduction

The BPB/MBE programme is now organized as a corpus of source-locked gates rather than as a single global curve fit. Paper 0 defines the dictionary. Paper I fixes the low-redshift MAP1782 branch and already requires the separation

$$\mu_m \neq \Sigma, \tag{1}$$

because matter growth and Weyl/lensing response cannot be represented by a universal scalar multiplier. Paper II shows that the galactic SPARC sector selects a derivative-like Δ_{kW^2} response and locks the active-galaxy sign rule. Paper III sharpens the CMB/Weyl hierarchy: the vanilla late-density dictionary is catastrophically excluded, while the strict face-minus dictionary survives Planck-lite, ACT lensing, and Planck lensing diagnostics with explicit guardrails. Paper IV propagates the strict branch to neutron-star structure as an effective compact-object tidal response. Paper V shows that early galaxies do not kill the branch once the fixed LCDM-like halo-mass UV pivot is replaced by a physical peak-host mapper. Paper VII, downstream of this one, fixes the acceleration scale a_0 and tests its BTFR and raw baryonic-rotation-curve consequences.

The present paper addresses the sixth obstruction: E_G.

The E_G statistic is designed to compare lensing, galaxy clustering, and velocity growth. It sits exactly at the interface between matter response and Weyl response. A naive one-Σ interpretation would demand that ACT lensing and E_G measure the same object. They do not. ACT lensing is primarily a Weyl auto-power probe. E_G is a Weyl–matter or lensing–galaxy cross-response probe. The central claim of Paper VI is therefore

$$\mu_m \neq \Sigma_{\rm auto} \neq \Sigma_{\rm cross}. \tag{2}$$

2 Corpus placement and release navigation

This paper is part of the BPB/MBE corpus release. The release is navigated by the root-level files

```
00_README_CORPUS
00_MANIFEST_CORPUS.csv
00_CLAIMS_INDEX.csv
00_REVIEWER_ROUTE
```

which separate human orientation, machine-readable file inventory, claim-to-source mapping, and referee quick-route review. Paper VI should be read as the E_G/Weyl–matter cross-response gate in that corpus, not as a standalone final model-selection claim.

Table 1: Paper VI placement in the BPB/MBE corpus.

Layer	Role	Paper VI contribution
Paper 0	Dictionary	Supplies $a_{\rm bg}$ and sector separation
Paper I	Low-z MAP	Establishes $\mu_m \neq \Sigma$ at joint-likelihood level
Paper III	CMB/Weyl	Establishes Weyl auto-sector support
Paper VI	E_G cross response	Separates $\Sigma_{\rm auto}$ from $\Sigma_{\rm cross}$
Reviewer route	Audit map	Points directly to the k-window, E_G, and projector gates

3 Why E_G is the stress test

In schematic continuum notation, E_G compares the Weyl potential to the growth field,

$$E_G(z) \sim \frac{\nabla^2(\Phi + \Psi)}{f(z)\delta}, \tag{3}$$

or, observationally, a ratio involving a lensing–galaxy cross-correlation, a galaxy auto-correlation, and a redshift-space distortion estimate of $\beta = f/b$ [1, 2, 3]. It is therefore dangerous for any modified-gravity model that treats lensing and matter growth as separable responses only in name.

For BPB/MBE this danger is also an opportunity. Papers I and III already show that the model is not a one-function deformation of LCDM. The primary-CMB dictionary, the matter-growth response μ_m, and the Weyl auto-response $\Sigma_{\rm auto}$ are distinct projections. E_G adds one more projection: the Weyl–matter cross response.

4 Auto versus cross Weyl sectors

We write the auto-Weyl power measured by ACT/Planck lensing as

$$P_{WW}(k, z) = \Sigma^2_{\rm auto}(k, z) P_{\delta\delta}(k, z). \tag{4}$$

The Weyl–matter cross-power relevant for E_G is written as

$$P_{W\delta}(k, z) = r_{W\delta}(k, z)\, \Sigma_{\rm auto}(k, z) P_{\delta\delta}(k, z). \tag{5}$$

This defines an effective cross response

$$\Sigma_{\rm cross}(k,z) = r_{W\delta}(k,z)\Sigma_{\rm auto}(k,z). \tag{6}$$

If all enhanced Weyl auto-power were perfectly traced by galaxies and velocities, $r_{W\delta} = 1$. If the Weyl auto-sector contains a component not fully correlated with the observed galaxy/RSD field, then

$$0 < r_{W\delta} < 1. \tag{7}$$

This is the minimal mathematical distinction required by the observables.

For the current compressed diagnostic, the scalar values are

$$r_{W\delta}^{\rm free} = 0.633563, \qquad \Sigma_{\rm auto} = 1.028105, \tag{8}$$

and hence

$$\Sigma_{\rm cross} = r_{W\delta}\Sigma_{\rm auto} \simeq 0.65137. \tag{9}$$

Figure 1 displays the response hierarchy and the projected closure $\sqrt{a_{\rm bg}}$.

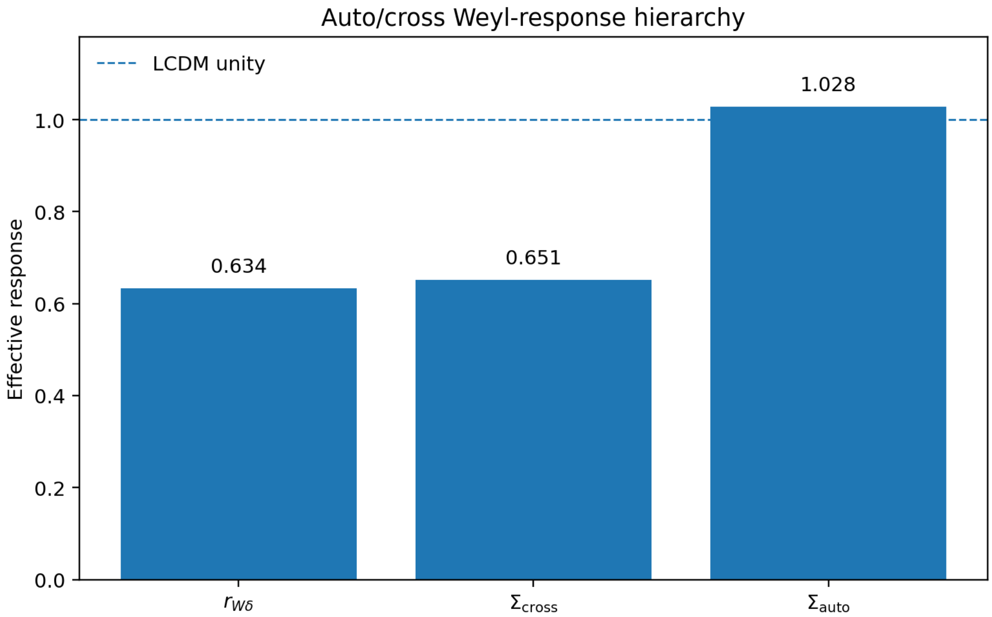

Figure 1: Compressed response hierarchy. ACT/Planck lensing constrains the Weyl auto-response $\Sigma_{\rm auto}$, while E_G is sensitive to $\Sigma_{\rm cross} = r_{W\delta}\Sigma_{\rm auto}$. The v0.2 projected closure $\sqrt{a_{\rm bg}} = 0.635364$ is visually indistinguishable from the free compressed $r_{W\delta} = 0.633563$ at this scale.

5 Corrected scale diagnostic: no ultra-low-k bin

The v0.0.0 draft displayed a k-window table that was not consistent with the simple convention $k = \ell/\chi$ for the stated ℓ and χ values. The v0.1 correction, retained here, is to either specify a more complex $k_{\rm eff}$ convention or use the simple geometric diagnostic explicitly. We choose the latter.

Using $k = \ell/\chi$ gives Table 2.

Table 2: Simple $k = \ell/\chi$ scale diagnostic. The conclusion is convention-robust: neither bin lies in the ultra-low-$k < 0.01$ or $k < 0.02\,\mathrm{Mpc}^{-1}$ regime.

Bin	$z_{\rm eff}$	$\ell_{\rm min}$	$\ell_{\rm max}$	χ [Mpc]	$k_{\rm min}$	$k_{\rm max}$
LOWZ	0.316	48	233	1290	0.0372	0.1805
CMASS	0.555	48	420	2120	0.0226	0.1980

The important result is not the last decimal of k. It is that all relevant bins satisfy

$$k > 0.02\,\mathrm{Mpc}^{-1}. \tag{10}$$

Therefore the earlier ultra-low-k explanation is not physically viable for Wenzl-like ACT+BOSS E_G bins. A transition requiring $k \lesssim 0.01\,\mathrm{Mpc}^{-1}$ cannot explain these data.

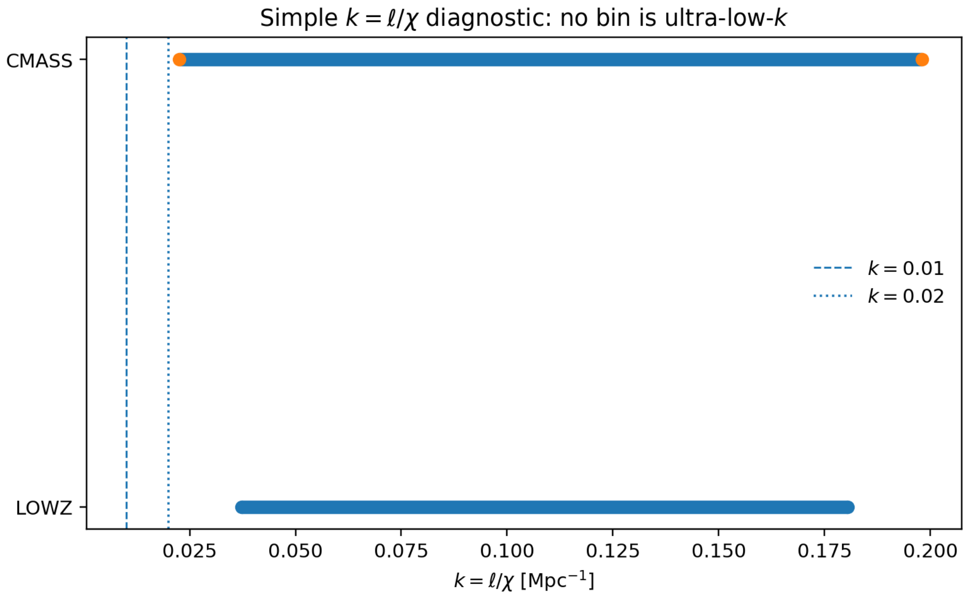

Figure 2: Simple $k = \ell/\chi$ diagnostic. The dashed and dotted vertical lines mark $k = 0.01$ and $0.02\,\mathrm{Mpc}^{-1}$. Both LOWZ and CMASS lie above the ultra-low-k regime.

6 Compressed E_G closure

Using the MAP1782 branch and the ACT-motivated Weyl auto closure, the minimal projected diagnostic gives

$$r_{W\delta}^{\mathrm{free}} = 0.633563, \qquad \chi^2_{E_G} = 0.191532. \tag{11}$$

The two-bin comparison is given in Table 3.

Table 3: Compressed E_G diagnostic. Residuals are quoted in units of the observational standard deviation.

Bin	Observation	σ	BPB prediction	Residual
LOWZ	0.430	0.100	0.3918	-0.382σ
CMASS	0.340	0.050	0.3507	$+0.214\sigma$

Using rounded residuals gives $\chi^2 \simeq 0.1917$, consistent with the source value 0.191532 within display precision. Both bins sit below 0.4σ in absolute residual.

This very small value should be interpreted with the compression and parameter counting in mind. The diagnostic has two compressed BOSS bins and one effective cross-response parameter, $r_{W\delta}$, in the free check. Therefore the small χ^2 is not advertised as a high-dimensional goodness-of-fit or as evidence that the covariance is perfect. It says that the two compressed residuals selected by the auto/cross closure are mutually consistent, and the stronger test is that the independently imported projected closure $r_{W\delta} = \sqrt{a_{\mathrm{bg}}}$ gives a fixed-minus-free penalty of only $\Delta\chi^2 = 0.008545$. A full assessment of underestimated errors, galaxy-bias nuisance, and window covariance belongs to the next projected-likelihood gate.

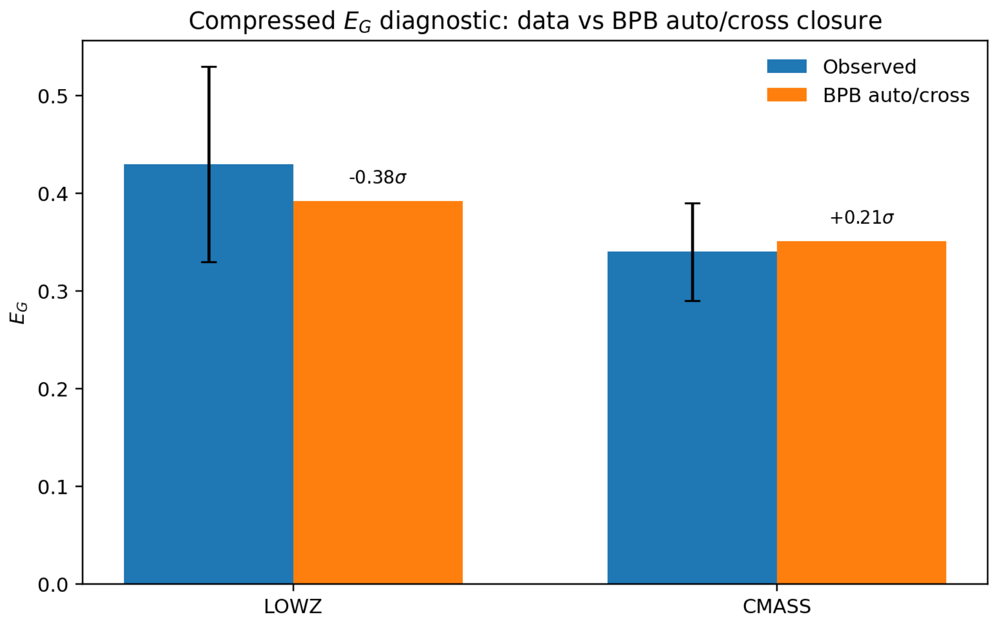

Figure 3: Compressed E_G observations compared with the BPB auto/cross prediction. Both LOWZ and CMASS residuals are below 0.4σ.

7 Density-weighted projected Weyl–matter closure

The compressed E_G diagnostic introduced an effective Weyl–matter correlation coefficient $r_{W\delta}$. The v0.2 closure is that the free value is reproduced by the background-dictionary projector

$$r_{W\delta} = \sqrt{a_{\rm bg}} = 0.635364421415, \qquad a_{\rm bg} = 0.403687948. \tag{12}$$

This is not an E_G fit. The value $a_{\rm bg}$ is inherited from the BPB background/density-weighted dictionary.

A useful way to interpret this closure is through a two-sector Weyl mixing angle. Let the effective sector weights entering the scalar Weyl response be

$$\rho_g^{\rm eff} : \rho_f^{\rm eff} = a_{\rm bg} : (1 - a_{\rm bg}). \tag{13}$$

If the two scalar Weyl modes obey the density-weighted mixing relation

$$\tan^2\theta_W = \frac{\rho_f^{\rm eff}}{\rho_g^{\rm eff}}, \tag{14}$$

then

$$\tan^2\theta_W = \frac{1 - a_{\rm bg}}{a_{\rm bg}}. \tag{15}$$

It follows algebraically that

$$\cos^2\theta_W = \frac{1}{1 + \tan^2\theta_W} = \frac{1}{1 + (1 - a_{\rm bg})/a_{\rm bg}} = a_{\rm bg}, \tag{16}$$

and therefore

$$r_{W\delta} = \cos\theta_W = \sqrt{a_{\rm bg}}. \tag{17}$$

The corresponding angle is

$$\theta_W = 50.552981835172^\circ, \qquad \frac{\rho_f^{\rm eff}}{\rho_g^{\rm eff}} = 1.477160898546. \tag{18}$$

Table 4: Density-weighted projected Weyl–matter closure values. The closure is conditional on Eq. (14); the explicit HR/BPB perturbation-equation source match remains an open guardrail.

Quantity	Value
$a_{\rm bg}$	0.403687948
$1 - a_{\rm bg}$	0.596312052
$\sqrt{a_{\rm bg}}$	0.635364421415
$\rho_f^{\rm eff}/\rho_g^{\rm eff}$	1.477160898546
θ_W	50.552981835172 deg
free $r_{W\delta}$ diagnostic	0.633563
$\Delta\chi^2$ fixed-minus-free	0.008545

This closure is non-circular at the projected diagnostic level: $a_{\rm bg}$ is inherited from the BPB background dictionary, not fitted to E_G. The E_G auto/cross diagnostic then tests the consequence. In the amplitude-space artefact, the audited relation gives

$$\frac{\langle X, X\rangle}{\langle A, A\rangle} = 0.397990327336, \qquad \sqrt{\frac{\langle X, X\rangle}{\langle A, A\rangle}} = 0.630864745675. \tag{19}$$

The squared value differs from $a_{\rm bg}$ by only -1.411392%, with all-row compressed support $\chi^2/N = 0.226490429325$.

The same gate also records a false route that must not be used as proof. The Wenzl 19-bin projected shape/model space is effectively one-dimensional, with dominant trace fraction about 0.9999987 unweighted and 0.9999989 ℓ-width weighted. Therefore a projector constructed inside that shape space is not evidence for the $a_{\rm bg}$ closure; it is an obstruction diagnostic. The response information must live in the auto/cross amplitude space or in the underlying raw Weyl response, not in the nearly collinear 19-bin shape basis.

Finally, the D3b s_0-weighted scalar-projector audit is supportive but not exact. The condition $(y_t^2/s_0)(1-a) = a^3$ predicts $a = 0.405838$ using the rounded $s_0 = 0.800$ value and $a = 0.406433$ using the exact $s_0 = 0.7956924397708606$, corresponding to relative offsets of 0.532556% and 0.680002% from the locked $a_{\rm bg}$. This is recorded as sub-percent support, not as an exact identity.

8 Physical interpretation

The interpretation is direct:

> ACT measures Weyl auto-power enhancement. E_G measures the part of Weyl structure correlated with galaxies and velocities.

If BPB/MBE produces an enhanced Weyl auto-sector, but only a fraction of that sector is matter-correlated in the galaxy/RSD projection, then ACT and E_G can both be satisfied.

This is not a loss of predictivity. It is an observable-sector distinction. Forcing primary-CMB response, matter growth, Weyl auto-power, Weyl–matter cross-power, neutron-star tidal response, high-redshift galaxy mapping, and BTFR acceleration into one scalar parameter creates artificial contradictions. The corpus instead identifies which response sector each observable probes.

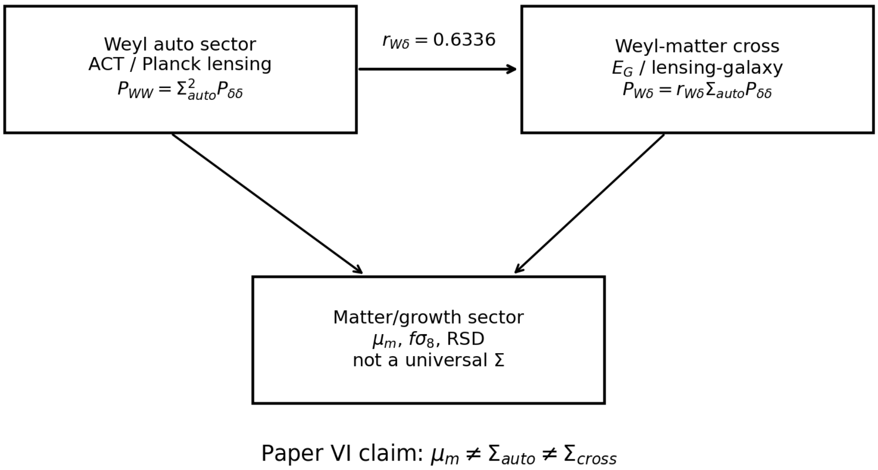

Figure 4: Auto/cross response schematic. The Weyl auto-sector controls lensing auto-power, while E_G probes the Weyl–matter cross-sector.

9 What v0.2 improves over v0.1

The v0.1 upgrade corrected the k-window table, made the auto/cross distinction explicit, and included figures/source CSVs for the compressed E_G comparison, the k diagnostic, the response hierarchy, and the schematic.

The v0.2 upgrade adds four items.

1. It inserts the projected density-weighted closure $r_{W\delta} = \sqrt{a_{\rm bg}}$.
2. It records that $a_{\rm bg}$ is imported from the BPB background dictionary rather than fitted to E_G.
3. It explicitly forbids the Wenzl shape-space projector as proof, because the projected response space is nearly one-dimensional.
4. It records the D3b s_0-weighted sub-percent support while keeping the exact field-equation derivation as a guardrail.

10 What this paper does not yet claim

This v0.2.1 paper remains a diagnostic closure. It does not yet compute the full Wenzl likelihood from first principles. It does not yet evaluate the full projected spectra

$$C_\ell^{\kappa g}, \qquad C_\ell^{gg}, \qquad \beta(z), \tag{20}$$

with survey windows, covariance, RSD treatment, galaxy bias, masks, and nuisance profiling.

It also does not claim that Eq. (14) has been derived from a complete action-level HR/BPB perturbation calculation. The internal source-match audit did not yet find a non-circular derivation of that density-weighted scalar/Weyl mixing relation in the existing HR/BPB perturbation files. Thus the allowed statement is a conditional density-weighted projected closure, not a final field-equation proof.

11 Conclusions

1. E_G is the natural sixth BPB/MBE gate because it sits at the boundary between Weyl lensing, galaxy clustering, and velocity growth.

2. A universal Weyl response Σ is not sufficient. Paper VI sharpens the hierarchy to

$$\mu_m \neq \Sigma_{\text{auto}} \neq \Sigma_{\text{cross}}. \tag{21}$$

3. The simple ultra-low-k explanation fails. With the explicit $k = \ell/\chi$ diagnostic, LOWZ probes $k = 0.0372$–$0.1805\,\text{Mpc}^{-1}$ and CMASS probes $k = 0.0226$–$0.1980\,\text{Mpc}^{-1}$.

4. The minimal auto/cross response model distinguishes $P_{WW} = \Sigma_{\text{auto}}^2 P_{\delta\delta}$ from $P_{W\delta} = r_{W\delta}\Sigma_{\text{auto}} P_{\delta\delta}$.

5. The compressed free diagnostic gives $r_{W\delta}^{\text{free}} = 0.633563$ and $\chi^2_{E_G} = 0.191532$; this low value is interpreted as a two-bin compressed consistency check, not as a full projected-likelihood goodness-of-fit.

6. The projected density-weighted closure gives $r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364421415$, with fixed-minus-free penalty $\Delta\chi^2 = 0.008545$.

7. The predictions LOWZ $= 0.3918$ and CMASS $= 0.3507$ are consistent with 0.430 ± 0.100 and 0.340 ± 0.050.

8. Therefore E_G does not presently refute BPB/MBE. It identifies the missing response distinction: ACT sees Weyl auto-power, while E_G sees Weyl–matter cross-power.

9. The next gate is a full projected $C_\ell^{\kappa g}$, C_ℓ^{gg}, β, and covariance likelihood.

A Claim source lock

Table 5: Paper VI v0.2 claim source lock. Full machine-readable rows are in `data/paperVI_v0_2_claim_source_lock.csv`.

Claim	Level	Locked value
P6.V02.01	AUTO/CROSS	$\mu_m \neq \Sigma_{\text{auto}} \neq \Sigma_{\text{cross}}$; $\Sigma_{\text{auto}} = 1.028105$, $\Sigma_{\text{cross}} = 0.65137$
P6.V02.02	K WINDOW	LOWZ $k = [0.0372, 0.1805]$, CMASS $k = [0.0226, 0.1980]$ Mpc^{-1}
P6.V02.03	COMPRESSED E_G	$r_{W\delta}^{\text{free}} = 0.633563$, $\chi^2 = 0.191532$
P6.V02.04	PROJECTOR	$r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364421415$, $\Delta\chi^2 = 0.008545$
P6.V02.05	FALSE ROUTE	Wenzl shape-space projector forbidden as proof; trace fraction $\simeq 0.999999$
P6.V02.06	AMPLITUDE SUPPORT	$\langle X, X\rangle/\langle A, A\rangle = 0.397990327336$, relative offset -1.411392%
P6.V02.07	D3B SUPPORT	s_0-weighted diagnostic within 0.53–0.68%, not exact
P6.V02.08	GUARDRAIL	No full projected likelihood or action-level perturbation proof claimed

B Source and reproducibility inventory

The v0.2.1 source package contains the figure/data source locks

```
data/eg_compressed_v0_1.csv
data/k_windows_v0_1.csv
data/response_hierarchy_v0_1.csv
data/paperVI_v0_2_numeric_locks.csv
data/paperVI_v0_2_claim_source_lock.csv
data/paperVI_v0_2_decisions.csv
scripts/make_paperVI_v0_2_figures.py
```

The corpus release navigation files are included under `release_index/`.

C Audit status

The v0.0.0 audit status was 23/27. The four real flags concerned the old k-window numbers. They were fixed in v0.1 by replacing the ambiguous window values with the explicit $k = \ell/\chi$ diagnostic. The central E_G closure numbers were already coherent: rounded residuals reproduce $\chi^2_{E_G} \simeq 0.1917$, consistent with the source value 0.191532, and the cross-response hierarchy satisfies $\Sigma_{\rm cross} = r_{W\delta}\Sigma_{\rm auto} \simeq 0.651 < \Sigma_{\rm auto} \simeq 1.028$.

The v0.2 audit adds the density-weighted projected closure and records the remaining D2 guardrail. The v0.2.1 hotfix adds the explicit two-bin/one-parameter interpretation of the small compressed E_G chi-square. This strengthens Paper VI without promoting it to a full projected likelihood or action-level proof.

References

[1] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, "Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity," *Physical Review Letters* **99**, 141302 (2007).

[2] R. Reyes et al., "Confirmation of general relativity on large scales from weak lensing and galaxy velocities," *Nature* **464**, 256–258 (2010).

[3] C. Blake et al., "RCSLenS: testing gravitational physics through the cross-correlation of weak lensing and large-scale structure," *Monthly Notices of the Royal Astronomical Society* **456**, 2806–2828 (2016).

[4] F. J. Qu et al., "The Atacama Cosmology Telescope: a measurement of the DR6 CMB lensing power spectrum and its implications for structure growth," *The Astrophysical Journal* **962**, 112 (2024).

[5] M. S. Madhavacheril et al., "The Atacama Cosmology Telescope: DR6 gravitational lensing map and cosmological parameters," *The Astrophysical Journal* **962**, 113 (2024).